NeubAItics Tech Pvt. Ltd.

SRM INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

INSTITUTION'S INNOVATION COUNCIL
(Ministry of HRD Initiative)

SRM Institute of Science and Technology
School of Mechanical Engineering
Department of Automobile Engineering

Kattankulathur - 603 203, Tamil Nadu, India

Organises

Hands-on Workshop on

Next Gen Vehicles: IOT, GPS and Autonomous Driving Technologies

Dates: 26th & 27th March 2026

Venue: Automotive electronics lab, First Floor, Automobile Engineering Department

Convenor:
Dr. G. Naresh

Co-Convenors:
Dr. T. Praveen Kumar
Dr. C. Carunaiselvane
Dr. S. Devakirubakaran

9940177837, 8122814437

NAAC	NMRD Govt.of India / UGC University Grants Commission	nirf NATIONAL INSTITUTIONAL RANKING FRAMEWORK	QS	THE WORLD UNIVERSITY RANKINGS	SHANGHAI RANKING	nature INDEX	UI GreenMetric World University Rankings
A++	Category 1 with 12B Status	(2025) Ranked 11th University	(2026) World Ranking one among 54 Indian Universities	(2025) World Ranking one among 91 Indian Universities	(2025) World Ranking Ranked 8-9 in Indian Universities	(2025) World Ranking one among 18 Indian Universities	(2024) World Ranking one among 3 Indian Universities

CHIEF PATRONS

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Dr. Ravi Pachamuthu, Pro Chancellor, (Administration), SRMIST

Dr. P. Sathyanarayanan, Pro Chancellor, (Academic), SRMIST

Dr. R. Shivakumar, Vice President, SRMIST

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Dr. M. Cheralathan, Chairperson, School of Mechanical Engineering, SRMIST

Dr. K. Kamalakkannan, HoD, Department of Automobile, SRMIST

ORGANIZING COMMITTEE

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Refreshments and Networking Lunch will be provided. No accommodation

About the workshop:

This hands-on workshop introduces the fundamental principles of robotics, autonomous systems, and IoT-enabled machines through an advanced line follower robot platform. The program demonstrates how mobile robots perceive their environment, make motion-related decisions, and perform autonomous navigation using motors, motor drivers, PWM-based speed control, and multiple sensors.

Advanced features such as LiDAR-based obstacle detection, sensor prioritization, and remote monitoring are incorporated to connect the robotic system with real-world autonomous vehicle applications. The workshop emphasizes mechanical motion, actuator behavior, and system-level automation, providing practical exposure while avoiding software-intensive or purely theoretical approaches.

Expected Outcomes

- Understanding of mobile robot motion as a mechanical process
- Practical knowledge of motors, motor drivers, and PWM speed control
- Hands-on exposure to sensor-based autonomous behavior
- Awareness of LiDAR-based perception and safety systems
- Understanding of how autonomous vehicles combine multiple sensors
- Ability to relate a simple robot to real-world autonomous systems
- Improved confidence in automation and robotics fundamentals

Target Audience / Beneficiaries

- UG students in Mechanical Engineering
- PG students in Automation, Mechatronics, and Manufacturing disciplines
- Mechanical students seeking practical exposure to robotics and autonomy
- Faculty members planning to introduce robotics-based lab modules
- Institutions aiming to provide foundational autonomous systems training

ABOUT THE INSTITUTION:

SRM Institute of Science and Technology (Formerly known as SRM University) is one of the top-ranking Institution and the most preferred engineering destination in India. It is established in 1985 by the Founder Chancellor Dr. T. R. Paarivendhar. SRM Institute of Science and Technology (SRMIST) is functioning in four campuses located at Kattankulathur, Vadapalani and Ramapuram in Tamil Nadu and at Modi Nagar, Ghaziabad with over 50,000 students and 3,020 faculty members, offering wide range of Under Graduate, Post Graduate and Doctoral programs in Engineering, Management, Medicine & Health Sciences, Law and Science & Humanities. The Institution has moved up through international alliances and collaborative initiatives to achieve global excellence. Over 150 students have been sponsored to 35 foreign Universities like MIT, Carnegie Mellon, UC Davis, Warwick and Western Australia. Now the Institute enjoys an unsurpassed reputation in academic and corporate circles being the preferred manpower source for vision to be recognized as a world - class learning institution. SRMIST has been categorized as grade 'A' Institution by Ministry of Human Resource Development (MHRD); Government of India. SRMIST is accredited by NAAC with 'A++' Grade in the year 2024.

ABOUT THE DEPARTMENT:

Department of Automobile Engineering was established in 2004 with a mission, to prepare highly competent Automobile Engineers for global automotive industries through innovative industry-oriented curriculum and intensive practical training. The department offers four Undergraduate programs: B.Tech in Automobile Engineering, Automobile Engineering with Specialization in Automotive Electronics, Automotive Engineering, Automobile Engineering with Specialization in Vehicle Testing and three Post graduation programs: M.Tech. in Electric Vehicle Technology, Automotive Hybrid Systems Engineering and Automotive Technology.

Highly qualified and dynamic young faculty members shape up the millennials for the future. Well-equipped and State-of-the-art laboratory facilities such as Automotive Components lab, Vehicle Testing lab, Vehicle Dynamics lab, Machine Shop, Automotive Design lab, Automotive Electrical system lab, Automotive Electronics lab, Electric Vehicle lab, Driving Simulator lab and IC Engine lab are in place.

The Department of Automobile Engineering at Kattankulathur campus is affiliated to Automotive Skills Development Council (ASDC), Government of India. Major areas of research interest are Automotive Materials, Alternative fuels and Emission Control, Green Vehicle Technologies, Vehicle Dynamics and Automotive Design.

The Department is coordinating Seven student Teams of our university - The Conrods (BAJA SAEINDIA), Camber Racing (SUPRA SAEINDIA), 4ZE Racing (Formula Electric), Team Enigmatic Plodder (Quad Bike), Team Infeon Super Mileage (Shell Eco Marathon Asia), Team Full Throttle (Go-Kart Championship) and The TorQ (Tractor Design Competition). These teams have won several awards, rewards and recognitions by participating in National and International events.

Academic/Research Collaboration with ARAI, VALEO, Apollo Tyres. Research collaboration with ATALON, PPG ASIAN Paints, Mahindra Research Valley, Volvo India, Renault Nissan Technology Business Center India. Ford, Mahindra & Mahindra, TVS Motors, Hyundai, Royal Enfield & Toyota Kirloskar India, Stellantis, BMW have sponsored vehicles and components to various Labs.

PROGRAMS OFFERED:

Under Graduate

- B. Tech Automobile Engineering
- B. Tech Automobile Engineering *(Spl.in Automotive Electronics)*
- B. Tech Automotive Engineering *(Spl.in Electric & Hybrid Vehicles- ARAI, Pune)*
- B. Tech Automobile Engineering *(Spl. in Vehicle Testing- GARC, Chennai)*
- B. Tech Automobile Engineering *(Weekend)*

Post Graduate

- M. Tech Automotive Technology *(ARAI, Pune)*
- M. Tech Electric Vehicle Technology
- M.Tech in Automotive Hybrid Systems Engineering *(Regular & Weekened)*

Two-Day Workshop Schedule

Advanced Line Follower Robot with Autonomous Enhancements

Day 1 – Fundamentals and Core Robot Build

Forenoon Session - 9:30 AM – 12:30 PM

- Introduction to mobile robots and automation flow
- DC motor basics for mobile robots
- Speed control architectures
- PWM fundamentals and duty cycle control
- Power electronics basics for motor driving
- Motor driver fundamentals
- Basic programming concepts for motion control

Afternoon Session - 1:30 PM – 5:00 PM

- Sensor types used in mobile robots
  - IR sensors
  - LDRs and basic comparison
- Chassis assembly and wheel alignment
- Motor mounting and wiring through drivers
- Sensor placement and integration
- Programming logic for line detection and correction

Hands-On Activity

- Complete assembly of the line follower robot
- Upload and test line-following program
- Robot runs on a defined route map
- Robot successfully follows the given path autonomously

Day 2 – Advanced Sensors and Autonomous Behavior

Forenoon Session - 9:30 AM – 12:30 PM

- Limitations of basic line follower robots
- Introduction to advanced industrial sensors
- LiDAR working principle and environment sensing
- Obstacle detection and safety logic
- Overview of communication technologies (LoRa, GPS, WiFi, Bluetooth)
- Basics of wireless control and monitoring

Afternoon Session - 1:30 PM – 5:00 PM

- Integration of LiDAR with line follower robot
- Sensor priority and basic sensor fusion
- Autonomous decision making logic
- Wireless start, stop, or status monitoring
- Debugging and system-level problem solving

Hands-On Challenge Session

- Convert basic line follower into an autonomous mobile robot
- Robot follows line while avoiding obstacles using LiDAR
- Students solve real-time challenges such as
  - Obstacle on path
  - Speed adjustment zones
  - Sensor conflict handling

SPEAKERS

Mr. Murugesan N
IoT Engineer

Mrs. Vasumathi R
Head of Robotics and IoT Department

Bank details:

A/C. No	: 117001001511846
A/C Holder Name	: 4ZE RACING
Bank Name	: CITY UNION BANK
IFSC CODE	: CIUB0000117
Branch	: CHENNAI TAMBARAM

Workshop fee details:

Industrialists	: Rs 2500
Student / Research Scholar	: Rs 2000

(Note: Rs 500 concession applicable for SRMIST participants)

REGISTRATION DETAILS:

Registration Link: https://forms.gle/tFVLhkG9umEMutuFA

SCAN HERE!